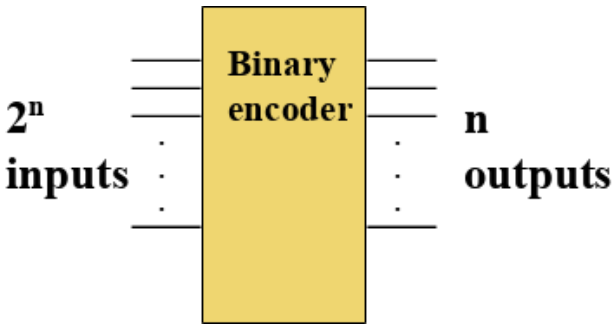
Combinational Logic - Encoders

5. Binary Encoders

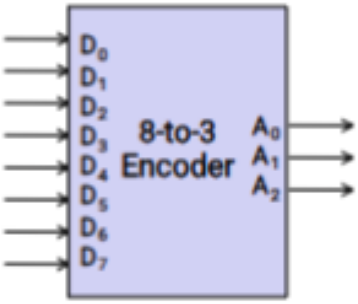
Encoding is the exact opposite of decoding. An encoder has 2^n input lines and n output lines.

- It assumes that **exactly one** input line is logic 1 at any given time.
- It outputs the binary code corresponding to the physical position (subscript number) of the active input.



8-to-3 Encoder Truth Table

- 2^3 = 8 inputs, 3 outputs
- one input =1, others = 0's
- Each input generate unique binary code



D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	A_2	A_1	A_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

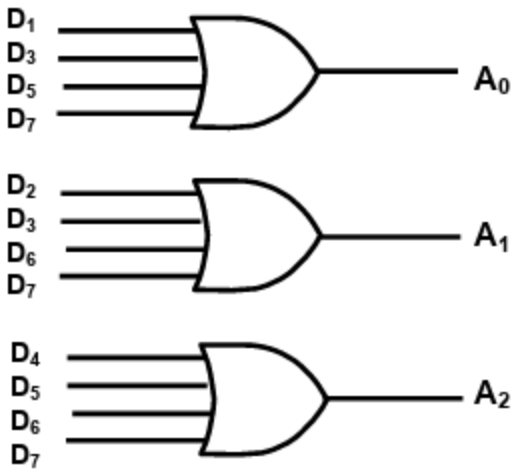
Derived Boolean Expressions:

By looking at the columns for A_0, A_1, and A_2, you can derive the equations by ORing together the specific D inputs that produce a `1` in that respective output column:

$$A_0 = D_1 + D_3 + D_5 + D_7$$

$$A_1 = D_2 + D_3 + D_6 + D_7$$

$$A_2 = D_4 + D_5 + D_6 + D_7$$



Standard encoders have two fatal flaws:

1. **Multiple Active Inputs:** If two inputs are active simultaneously (e.g., $D_3 = 1$ and $D_6 = 1$), the output is garbled (it outputs `111`, which is incorrect for both).
2. **All Zero Inputs:** If all inputs are 0, the output is `000`. However, `000` is also the output when D_0 is active! The system cannot tell the difference between "Input 0 is active" and "The machine is turned off".

6. The Priority Encoder

The Priority Encoder is designed to solve the two limitations of standard encoders.

1. If multiple inputs are active, it only pays attention to the input with the **highest priority** (the highest subscript number).
2. It includes a third output called the **Valid Bit (V)**. If any input is active, $V = 1$. If all inputs are zero, $V = 0$.

4-to-2 Priority Encoder (Condensed Truth Table): Because we ignore lower-order bits when a high-priority bit is active, we use "Don't Cares" (X) to condense a 16-row truth table down to just 5 rows.

D_3	D_2	D_1	D_0	A_1	A_0	V
0	0	0	0	X	X	0
0	0	0	1	0	0	1
0	0	1	X	0	1	1
0	1	X	X	1	0	1
1	X	X	X	1	1	1

(Observe how D_3 overrides everything below it).

K-Map Simplified Equations: Applying the condensed truth table to Karnaugh Maps yields the minimized hardware equations:

- **Output A0:** $A_0 = D_3 + D_1\overline{D_2}$
- **Output A1:** $A_1 = D_2 + D_3$
- **Valid Bit:** $V = D_0 + D_1 + D_2 + D_3$

